Data Assimilation with Stochastic Interpolants

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Abstract

Data assimilation combines mathematical models with observational data to estimate the state of complex dynamical systems. However, efficiently sampling from high-dimensional posterior distributions in nonlinear dynamical systems remains a major computational challenge. Here we show that stochastic interpolants can be adapted to perform exact posterior sampling for data assimilation without requiring model retraining. Our method derives a posterior stochastic differential equation incorporating observational information through drift correction, enabling flexible numerical solvers and tractable likelihood score computation via observation interpolation. Our framework provides superior uncertainty quantification while maintaining computational efficiency. We showcase the efficacy of the approach on series of test cases.

1 Introduction

Data assimilation (DA) addresses the fundamental challenge of estimating unknown states or parameters in dynamical systems by optimally combining prior model predictions with noisy observations. This is typically accomplished through either variational methods, which minimize regularized objective functions, or Bayesian methods, which estimate posterior distributions – usually via finite sample approximations Evensen et al. (2022). The Bayesian approach is often preferred due to its natural framework for uncertainty quantification through posterior distribution estimation.

Within Bayesian DA, approaches such as ensemble Kalman filters and particle filters are arguably the most popular. Ensemble-based Kalman filters are computationally efficient but require Gaussianity assumptions and can only handle weakly nonlinear forward and observation models. Particle filters, conversely, impose virtually no restrictions but are computationally expensive due to the large number of samples required. This fundamental trade-off between computational tractability and modeling flexibility has motivated extensive research into alternative approaches.

One promising direction replaces expensive forward models with efficient neural network surrogates Mücke et al. (2024); Cheng et al. (2022); Zhong et al. (2023); Chen et al. (2023). While achieving substantial computational speed-ups, these deterministic approximations require specifying a priori model error distributions when incorporated into DA frameworks. This poses a fundamental challenge for uncertainty quantification in high-dimensional stochastic systems, where the true model error characteristics are unknown.

Recently, generative models have emerged as a compelling alternative, learning to directly sample from distributions of interest. In DA contexts, these models are trained on prior

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distributions and subsequently modified to incorporate observational constraints for posterior sampling. Although frameworks like denoising diffusion models Ho et al. (2020) and flow matching models Lipman et al. (2023) have achieved remarkable success in computer vision, their iterative sampling procedures can be computationally demanding, potentially limiting real-time DA applications.

Contributions: We introduce a novel posterior sampling method based on stochastic interpolants (SIs), a recent class of generative models that transform samples between arbitrary distributions via stochastic differential equations. We extend this framework by deriving a posterior SDE that incorporates observational information through drift correction. Our key contributions are:

- We derive a posterior drift expression by incorporating likelihood scores into the SI framework without requiring retraining.
- We prove that the resulting SDE generates exact posterior samples under ideal conditions.
- We introduce a practical likelihood score estimation method using observation interpolations and derive a correction that compensates for approximation errors.

2 Related work

2.1 Flow-based generative models in physics applications

Flow-based generative models have received substantial attention in recent years due to their success in generating images and videos. A similar development is occurring in physics applications, where these models are trained to perform probabilistic forecasting by learning to sample from the conditional distribution of the next state given the current state. Several approaches have emerged to address this challenge. Some methods combine deterministic neural operators with noise-based techniques: Lippe et al. (2023) perform initial predictions using neural operators and add/remove noise inspired by denoising diffusion models, while Gao et al. Gao et al. (2024) employ neural networks for time-stepping in latent space followed by physics-informed diffusion upsampling. Other approaches focus on direct conditional sampling. Kohl et al. (2024) trains denoising diffusion models to sample directly from the next state, while Fotiadis et al. (2024) uses flow-matching models for direct conditional sampling. Chen et al. (2024) employs stochastic interpolant models to transform current states into samples from the conditional distribution, which Mücke & Sanderse (2025) extended by incorporating energy consistency into the training.

2.2 Inverse problems with generative models

Generative models are increasingly utilized for solving inverse problems, as detailed in the survey by Daras et al. (2024). For linear inverse problems, several training-free methods have been developed. Pokle et al. (2024) introduces a method for image inverses via flows, while Zhang et al. (2025) utilizes flow priors through iterative corrupted trajectory matching. Recent advancements also include Yan et al. (2024), which proposes Flow with Interpolant Guidance (FIG). In the domain of stochastic interpolation, Ahamed & Haber (2024) developed DAWN-SI for data-aware and noise-informed solutions, and Negrel et al. (2025) explored multitask learning.

2.3 Data assimilation with generative models

The integration of generative models into data assimilation (DA) seeks to overcome the limitations of traditional methods like ensemble Kalman filters and particle filters. Early efforts in this direction include score-based data assimilation by Rozet & Louppe (2023). For high-dimensional systems, Bao et al. (2024) introduced an ensemble score filter, which was further extended by Si & Chen (2024) to a latent space for handling sparse observation data. Multimodal settings have been addressed by Qu et al. (2024) through deep generative DA. Most relevant to our approach is FlowDAS, introduced by Chen et al. (2025), which utilizes a stochastic interpolant-based framework for DA.

3 Preliminaries

We denote discrete physical time steps with superscripts and pseudo-time steps with subscripts. E.g. $\boldsymbol{u}^n$ denotes a quantity at physical time step n and $\boldsymbol{x}_\tau$ denotes a quantity at pseudo-time τ. Furthermore, the notation $\boldsymbol{u}^{n:m}$, with $n < m$, refers to the time series $(\boldsymbol{u}^n, \boldsymbol{u}^{n+1}, \dots, \boldsymbol{u}^{m-1}, \boldsymbol{u}^m)$.

3.1 Bayesian data assimilation

We briefly outline the Bayesian data assimilation setting in this subsection. For more information on data assimilation, see Carrassi et al. (2018); Evensen et al. (2022).

We consider the discrete dynamical system:

$$\boldsymbol{u}^n = \boldsymbol{F}(\boldsymbol{u}^{n-1}) + \boldsymbol{\xi}^n, \qquad \boldsymbol{\xi}^n \sim p_{\boldsymbol{\xi}}, \tag{1a}$$

$$\boldsymbol{y}^n = \boldsymbol{H}(\boldsymbol{u}^n) + \boldsymbol{\eta}^n, \qquad \boldsymbol{\eta}^n \sim p_{\boldsymbol{\eta}}, \tag{1b}$$

where, for all $n = 1, \dots, N$, $\boldsymbol{u}^n \in \mathbb{R}^{N_{\boldsymbol{u}}}$ is the state, $\boldsymbol{y}^n \in \mathbb{R}^{N_{\boldsymbol{y}}}$ are the observations, $\boldsymbol{\xi}^n \in \mathbb{R}^{N_{\boldsymbol{u}}}$ is the model noise, $\boldsymbol{\eta}^n \in \mathbb{R}^{N_{\boldsymbol{y}}}$ is the observation noise, $\boldsymbol{F} : \mathbb{R}^{N_{\boldsymbol{u}}} \to \mathbb{R}^{N_{\boldsymbol{u}}}$ is the forward operator, and $\boldsymbol{H} : \mathbb{R}^{N_{\boldsymbol{u}}} \to \mathbb{R}^{N_{\boldsymbol{y}}}$ is the observation operator. The system can be the result of a discretized PDE system and is therefore potentially very high-dimensional. Eq. (1a) induces a prior transition density, $p(\boldsymbol{u}^n \mid \boldsymbol{u}^{n-1})$, that propagates the state without incorporating observations. Furthermore, the observation likelihood can be evaluated via the noise distribution, $p(\boldsymbol{y}^n \mid \boldsymbol{u}^n) = p_\eta(\boldsymbol{y}^n - \boldsymbol{H}(\boldsymbol{u}^n))$.

In this work, we are interested in the posterior distribution:

$$p(\boldsymbol{u}^{0:n} \mid \boldsymbol{y}^{1:n}) = p(\boldsymbol{u}^0) \prod_{i=1}^{n} p(\boldsymbol{u}^i \mid \boldsymbol{y}^i, \boldsymbol{u}^{i-1}) \,. \tag{2}$$

Eq. (2) holds due to the Markovian property of Eq. (1) Carrassi et al. (2018); Evensen et al. (2022). We can sample from this distribution by autoregressively sampling from the one-step posterior:

$$p(\boldsymbol{u}^n \mid \boldsymbol{y}^n, \boldsymbol{u}^{n-1}) = \frac{p(\boldsymbol{y}^n \mid \boldsymbol{u}^n)\, p(\boldsymbol{u}^n \mid \boldsymbol{u}^{n-1})}{p(\boldsymbol{y}^n)}. \tag{3}$$

Therefore, we focus on sampling from Eq. (3) for the rest of the paper.

3.2 Stochastic interpolants for probabilistic forecasting

We construct a generative model for probabilistic forecasting using the stochastic interpolant (SI) framework Albergo et al. (2023); Albergo & Vanden-Eijnden (2023); Chen et al. (2024). The objective is to generate samples from the forecast prior $p(\boldsymbol{u}^n \mid \boldsymbol{u}^{n-1})$ in Eq. (3) by constructing an SDE that transports samples from a base distribution to the target distribution in pseudo-time $\tau \in [0, 1]$.

Consider densities $p(\boldsymbol{x}_0)$ and $p(\boldsymbol{x}_1 \mid \boldsymbol{x}_0)$ with $\boldsymbol{x}_0, \boldsymbol{x}_1 \in \mathbb{R}^d$. The SI framework generates samples from $p(\boldsymbol{x}_1 \mid \boldsymbol{x}_0)$ by solving the SDE

$$\mathrm{d}\boldsymbol{X}_\tau = \boldsymbol{b}_\theta(\boldsymbol{X}_\tau, \boldsymbol{x}_0, \tau)\mathrm{d}\tau + \gamma_\tau \mathrm{d}\boldsymbol{W}_\tau, \qquad \tau \in [0, 1], \quad \boldsymbol{X}_0 = \boldsymbol{x}_0, \tag{4}$$

where $\boldsymbol{b}_\theta$ is a neural network approximating a drift and γ_τ is a prescribed diffusion coefficient.

The central object in the SI framework is the interpolant

$$\boldsymbol{I}_\tau := \boldsymbol{I}_\tau(\boldsymbol{x}_0, \boldsymbol{x}_1) = \alpha_\tau \boldsymbol{x}_0 + \beta_\tau \boldsymbol{x}_1 + \gamma_\tau \boldsymbol{W}_\tau, \tag{5}$$

where $\boldsymbol{W}_\tau := \sqrt{\tau}\,\boldsymbol{z}$ with $\boldsymbol{z} \sim \mathcal{N}(0, \boldsymbol{I})$. The scalar functions α_τ, β_τ, γ_τ satisfy the temporal boundary conditions $\alpha_0 = \beta_1 = 1$ and $\alpha_1 = \beta_0 = \gamma_1 = 0$, which guarantee that $\boldsymbol{I}_0 \sim p(\boldsymbol{x}_0)$ and $\boldsymbol{I}_1 \sim p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)$. The stochastic dynamics of the interpolant for given $\boldsymbol{x}_0$ and $\boldsymbol{x}_1$ are:

$$\mathrm{d}\boldsymbol{I}_\tau = \underbrace{\left(\dot{\alpha}_\tau \boldsymbol{x}_0 + \dot{\beta}_\tau \boldsymbol{x}_1 + \dot{\gamma}_\tau \boldsymbol{W}_\tau\right)}_{=:\ \boldsymbol{R}_\tau(\boldsymbol{x}_0, \boldsymbol{x}_1)} \mathrm{d}\tau + \gamma_\tau \mathrm{d}\boldsymbol{W}_\tau. \tag{6}$$

From the interpolant dynamics one can show that the SDE (4) has the same marginal law as $\boldsymbol{I}_\tau$, and thus transports $p(\boldsymbol{x}_0)$ to $p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)$, when the drift is given by the conditional expectation

$$\boldsymbol{b}(\boldsymbol{x}, \boldsymbol{x}_0, \tau) = \mathbb{E}\left[\boldsymbol{R}_\tau(\boldsymbol{x}_0, \boldsymbol{x}_1) \mid \boldsymbol{X}_\tau = \boldsymbol{x}\right], \tag{7}$$

where the expectation is over $\boldsymbol{x}_1 \sim p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)$ and $\boldsymbol{z} \sim \mathcal{N}(0, \boldsymbol{I})$. The neural network $\boldsymbol{b}_\theta$ is trained to approximate this drift by minimizing the mean-squared-error loss

$$\mathcal{L}(\theta) = \int_0^1 \mathbb{E}\left[\|\boldsymbol{b}_\theta(\boldsymbol{I}_\tau, \boldsymbol{x}_0, \tau) - \boldsymbol{R}_\tau\|_2^2\right] \mathrm{d}\tau, \tag{8}$$

where the expectation is over $\boldsymbol{x}_0$, $\boldsymbol{x}_1 \sim p(\boldsymbol{x}_0, \boldsymbol{x}_1)$, and $\boldsymbol{z}$, and τ is drawn uniformly from $[0, 1]$. Crucially, no SDE solve is required during training.

Once trained, new samples $\boldsymbol{X}_1 \sim p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)$ are generated by numerically solving (4), with the choice of solver and number of time steps offering a flexible accuracy–cost trade-off. Setting $\boldsymbol{x}_0 = \boldsymbol{u}^{n-1}$ and $\boldsymbol{x}_1 = \boldsymbol{u}^n$, the model learns to sample from the one-step forecast distribution $p\big(\boldsymbol{u}^n \,\big|\, \boldsymbol{u}^{n-1}\big)$, and arbitrarily long trajectories are obtained by iterating this procedure.

4 Methodology

In this section we will outline the proposed methodology for posterior sampling with stochastic interpolants.

Let $\boldsymbol{b}_\theta$ be trained such that solving Eq. (4) with initial condition $\boldsymbol{x}_0$ and a chosen γ_τ generates samples from $p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)$. The aim is then to sample from Eq. (3) by deriving a modified drift term, $\boldsymbol{b}^*_\theta$, such that the SDE:

$$\begin{aligned} \mathrm{d}\boldsymbol{X}^*_\tau &= \boldsymbol{b}^*_\theta(\boldsymbol{X}^*_\tau, \boldsymbol{x}_0, \boldsymbol{y}, \tau)\mathrm{d}\tau + \gamma_\tau \mathrm{d}\boldsymbol{W}_\tau, \\ \boldsymbol{y} &= \boldsymbol{H}(\boldsymbol{x}_1) + \boldsymbol{\eta}, \end{aligned} \tag{9}$$

with $\tau \in [0, 1]$ and $\boldsymbol{X}_0 = \boldsymbol{x}_0$ generates samples from the posterior distribution, $\boldsymbol{X}^*_{\tau=1} \sim p(\boldsymbol{x}_1 \,|\, \boldsymbol{y}, \boldsymbol{x}_0)$. The posterior drift term, $\boldsymbol{b}^*_\theta$, should ideally not require additional training, but simply rely on the pre-trained drift term, $\boldsymbol{b}_\theta$, and the observations, $\boldsymbol{y}$.

4.1 From prior drift to posterior drift

To obtain posterior samples rather than prior samples, we add a correction term to the drift:

$$\boldsymbol{b}^*_\theta(\boldsymbol{x}, \boldsymbol{x}_0, \boldsymbol{y}, \tau) = \boldsymbol{b}_\theta(\boldsymbol{x}, \boldsymbol{x}_0, \tau) + \gamma_\tau^2 \nabla_{\boldsymbol{x}} \log \underbrace{\mathbb{E}[p(\boldsymbol{y} \,|\, \boldsymbol{x}_1) \,|\, \boldsymbol{X}_\tau = \boldsymbol{x}, \boldsymbol{X}_0 = \boldsymbol{x}_0]}_{=\Phi^{\boldsymbol{y}}_\tau(\boldsymbol{x}, \boldsymbol{x}_0)}. \tag{10}$$

The added term is the diffusion-scaled likelihood score. It is the expected value of the of the observation likelihood at the terminal pseudo-time given the current state at pseudo-time τ. Intuitively, this term pushes the process towards areas with large observation likelihood, while the prior drift term balances that by pushing the process towards the prior distribution. In total, the process should then produce samples from the posterior distribution.

To show that the stated drift term does indeed generate posterior samples, we have the following theorem:

Theorem 4.1 (Posterior SDE)**.** *Given a drift term, $\boldsymbol{b}_\theta$, trained using Eq.* (8)*, such that solutions to the SDE in Eq.* (4) *at time $\tau = 1$, $\boldsymbol{X}_1$ are distributed according to $p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)$. Furthermore, given observations, $\boldsymbol{y}$, likelihood $p(\boldsymbol{y} \,|\, \boldsymbol{x}_1)$, and function $\Phi_\tau^{\boldsymbol{y}}(\boldsymbol{x}, \boldsymbol{x}_0) = \mathbb{E}[p(\boldsymbol{y} \,|\, \boldsymbol{x}_1) \,|\, \boldsymbol{X}_\tau = \boldsymbol{x}, \boldsymbol{X}_0 = \boldsymbol{x}_0]$, the SDE,*

$$\mathrm{d}\boldsymbol{X}_\tau^* = \boldsymbol{b}_\theta(\boldsymbol{X}_\tau^*, \boldsymbol{x}_0, \tau)\mathrm{d}\tau + \gamma_\tau^2 \nabla_{\boldsymbol{X}_\tau^*} \log \Phi_\tau^{\boldsymbol{y}}(\boldsymbol{X}_\tau^*, \boldsymbol{x}_0)\mathrm{d}\tau + \gamma_\tau \mathrm{d}\boldsymbol{W}_\tau, \quad \tau \in [0,1], \quad \boldsymbol{X}_0^* = \boldsymbol{x}_0, \tag{11}$$

generates solutions with terminal states, $\boldsymbol{X}_1^*$, distributed according to the posterior $p(\boldsymbol{x}_1 \,|\, \boldsymbol{y}, \boldsymbol{x}_0)$.

See Appendix A for a proof.

4.2 Approximating the likelihood score via observation interpolant

To evaluate the posterior drift, we must evaluate the likelihood score at the intermediate SDE states, $\boldsymbol{x}_\tau$. Evaluating the likelihood at pseudo-time $\tau = 1$ requires computing an integral:

$$\Phi_\tau^{\boldsymbol{y}}(\boldsymbol{x}_\tau, \boldsymbol{x}_0) = \mathbb{E}[p(\boldsymbol{y} \,|\, \boldsymbol{x}_1) \,|\, \boldsymbol{X}_\tau = \boldsymbol{x}_\tau, \boldsymbol{X}_0 = \boldsymbol{x}_0] = \int p(\boldsymbol{y} \,|\, \boldsymbol{x}_1)\, p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_\tau, \boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_1 \tag{12}$$

Computing this integral is prohibitively expensive, as it requires sampling from $p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_\tau, \boldsymbol{x}_0)$, which necessitates solving the SDE forward in pseudo-time. Furthermore, we also have to backpropagate through the trajectory in pseudo-time to compute the gradient with respect to $\boldsymbol{x}_\tau$ to obtain the likelihood score. We propose to overcome this issue by defining an interpolant for the observations, $\bar{\boldsymbol{y}}_\tau$:

$$\bar{\boldsymbol{y}}_\tau(\boldsymbol{x}_0, \boldsymbol{y}) = \alpha_\tau \boldsymbol{H}(\boldsymbol{x}_0) + \beta_\tau \boldsymbol{y}, \tag{13}$$

where α_τ and β_τ are defined such that:

$$\bar{\boldsymbol{y}}_0(\boldsymbol{x}_0, \boldsymbol{y}) = \boldsymbol{H}(\boldsymbol{x}_0), \quad \bar{\boldsymbol{y}}_1(\boldsymbol{x}_0, \boldsymbol{y}) = \boldsymbol{y}. \tag{14}$$

Then, the expected likelihood at pseudo-time τ can be computed by simply evaluating the likelihood:

$$\bar{\Phi}_\tau^{\bar{\boldsymbol{y}}_\tau}(\boldsymbol{x}_\tau, \boldsymbol{x}_0) = p_\tau(\bar{\boldsymbol{y}}_\tau \,|\, \boldsymbol{x}_\tau, \boldsymbol{x}_0) \tag{15}$$

With this formulation, the drift to be used for the posterior SDE is

$$\boldsymbol{b}_\theta^*(\boldsymbol{x}_\tau, \boldsymbol{x}_0, \bar{\boldsymbol{y}}_\tau, \tau) = \boldsymbol{b}_\theta(\boldsymbol{x}_\tau, \boldsymbol{x}_0, \tau) + \gamma_\tau^2 \nabla_{\boldsymbol{x}_\tau} \log \bar{\Phi}_\tau^{\bar{\boldsymbol{y}}_\tau}(\boldsymbol{x}_\tau, \boldsymbol{x}_0) \tag{16}$$

Since we are now computing the likelihood with the interpolated observations, we require the distribution of $\bar{\boldsymbol{y}}_\tau$ given $\boldsymbol{x}_\tau$:

Lemma 4.2 (Interpolated observation likelihood)**.** *Let the observation operator $\boldsymbol{H}$ be linear with matrix H, and let the observation noise be Gaussian, $\boldsymbol{\eta} \sim \mathcal{N}(0, R)$. Then the interpolated observation satisfies*

$$\bar{\boldsymbol{y}}_\tau = H\boldsymbol{x}_\tau + \bar{\boldsymbol{\eta}}_\tau, \qquad \bar{\boldsymbol{\eta}}_\tau := \beta_\tau \boldsymbol{\eta} - \gamma_\tau H \boldsymbol{W}_\tau, \tag{17}$$

and the mean and covariance of the conditional distribution, $p(\bar{\boldsymbol{y}}_\tau \,|\, \boldsymbol{x}_0, \boldsymbol{x}_\tau)$, are:

$$\mathbb{E}\left[\bar{\boldsymbol{y}}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0\right] = H\boldsymbol{x}_\tau - \gamma_\tau H \mathbb{E}\left[\boldsymbol{W}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0\right] = \bar{\mu}_\tau(\boldsymbol{x}_\tau, \boldsymbol{x}_0), \tag{18}$$

$$Cov(\bar{\boldsymbol{y}}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0) = \beta_\tau^2 R + \gamma_\tau^2 H\, Cov(\boldsymbol{W}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0)\, H^\top = \bar{\Sigma}_\tau(\boldsymbol{x}_\tau, \boldsymbol{x}_0). \tag{19}$$

Proof is provided in Appendix B. The conditional expectation and covriance of the Wiener process, can be computed directly from the trained drift term.

Lemma 4.3 (Conditional moments of the Wiener process)**.** *The conditional mean and covariance of $\boldsymbol{W}_\tau$ given $(\boldsymbol{x}_\tau, \boldsymbol{x}_0)$ can be expressed in terms of the trained drift $\boldsymbol{b}_\theta$ as follows:*

$$\mathbb{E}[\boldsymbol{W}_\tau \mid \boldsymbol{X}_\tau = \boldsymbol{x}, \boldsymbol{X}_0 = \boldsymbol{x}_0] = -\gamma_\tau \tau A_\tau(\beta_\tau \boldsymbol{b}_\theta(\boldsymbol{x}, \boldsymbol{x}_0, \tau) - c_\tau(\boldsymbol{x}, \boldsymbol{x}_0)) = \mu_\tau^{\boldsymbol{W}}(\boldsymbol{x}_\tau, \boldsymbol{x}_0), \tag{20}$$

where $c_\tau(\boldsymbol{x}, \boldsymbol{x}_0) = \dot{\beta}_\tau \boldsymbol{x} + (\beta_\tau \dot{\alpha}_\tau - \dot{\beta}_\tau \alpha_\tau)\boldsymbol{x}_0$, and $A_\tau = \left[\tau\gamma_\tau(\dot{\beta}_\tau \gamma_\tau - \beta_\tau \dot{\gamma}_\tau)\right]^{-1}$. Furthermore,

$$Cov(\boldsymbol{W}_\tau \mid \boldsymbol{X}_\tau = \boldsymbol{x}, \boldsymbol{X}_0 = \boldsymbol{x}_0) = \tau \boldsymbol{I} + \gamma_\tau^2 \tau^2\, A_\tau \left[\beta_\tau J_{\boldsymbol{b}_\theta}(\boldsymbol{x}, \boldsymbol{x}_0, \tau) - \dot{\beta}_\tau \boldsymbol{I}\right] = \Sigma_\tau^{\boldsymbol{W}}(\boldsymbol{x}_\tau, \boldsymbol{x}_0), \tag{21}$$

where $J_{\boldsymbol{b}_\theta}(\boldsymbol{x}, \boldsymbol{x}_0, \tau) = \nabla_{\boldsymbol{x}} \boldsymbol{b}_\theta(\boldsymbol{x}, \boldsymbol{x}_0, \tau)$ is the Jacobian of the trained drift with respect to $\boldsymbol{x}$.

A proof can be found in Appendix B.

4.2.1 Multiplicative correction of the likelihood score

We assume the interpolant likelihood is Gaussian,

$$\tilde{p}(\bar{\boldsymbol{y}}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0) := \mathcal{N}\big(\bar{\mu}_\tau(\boldsymbol{x}_\tau, \boldsymbol{x}_0),\, \bar{\Sigma}_\tau(\boldsymbol{x}_\tau, \boldsymbol{x}_0)\big)\,, \tag{22}$$

and that the true likelihood admits a Gaussian surrogate,

$$\Phi_\tau^{\boldsymbol{y}}(\boldsymbol{x}_\tau, \boldsymbol{x}_0) \approx \mathcal{N}(\boldsymbol{y};\, \mu_\tau(\boldsymbol{x}_\tau, \boldsymbol{x}_0),\, R)\,, \qquad \mu_\tau := H\,\mathbb{E}[\boldsymbol{x}_1 \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0], \tag{23}$$

whose mean is the Tweedie conditional expectation pushed through H and whose covariance is the raw observation noise R. A closed form for μ_τ that is free of the drift Jacobian, as well as an explicit expression for $\nabla_{\boldsymbol{x}_\tau}\mu_\tau$, are given in Lemma D.1 and Lemma D.2 in Appendix D.

Let $\bar{S}_\tau := \nabla_{\boldsymbol{x}_\tau} \log \bar{\Phi}_\tau^{\bar{\boldsymbol{y}}_\tau}$ and $S_\tau := \nabla_{\boldsymbol{x}_\tau} \log \Phi_\tau^{\boldsymbol{y}}$ denote the interpolant- and true-likelihood scores. Our main result is that, under the Gaussian assumptions above, the two are related by a closed-form *multiplicative* gain acting only in the observation subspace.

Theorem 4.4 (Multiplicative correction)**.** *Under the assumptions* (22)–(23)*,*

$$S_\tau = G_\tau(\boldsymbol{x}_\tau, \boldsymbol{x}_0)\,\bar{S}_\tau, \qquad G_\tau(\boldsymbol{x}_\tau, \boldsymbol{x}_0) = \boldsymbol{I} + \frac{\gamma_\tau^2}{\beta_\tau^2}\,\Sigma_\tau^{\boldsymbol{W}}(\boldsymbol{x}_\tau, \boldsymbol{x}_0)\,H^\top R^{-1} H, \tag{24}$$

with $\Sigma_\tau^{\boldsymbol{W}}$ the conditional Wiener covariance of Lemma 4.3. In particular $G_\tau - \boldsymbol{I}$ has rank at most N_y and vanishes on $\ker H$*.*

See Appendix D for the proof. Substituting (24) into the posterior drift (16) gives

$$\boldsymbol{b}_\theta^*(\boldsymbol{x}_\tau, \boldsymbol{x}_0, \boldsymbol{y}, \tau) = \boldsymbol{b}_\theta(\boldsymbol{x}_\tau, \boldsymbol{x}_0, \tau) + \gamma_\tau^2\, G_\tau(\boldsymbol{x}_\tau, \boldsymbol{x}_0)\,\bar{S}_\tau, \tag{25}$$

so the exact-surrogate posterior drift is obtained from the interpolant-score drift by a single application of a rank-N_y operator.

Corollary 4.5 (Jacobian-free posterior drift)**.** *Under the approximation $\Sigma_\tau^{\boldsymbol{W}} \approx \tau\boldsymbol{I}$ of Eq.* (21)*, the gain reduces to the state-independent form*

$$G_\tau \approx \boldsymbol{I} + \frac{\gamma_\tau^2 \tau}{\beta_\tau^2}\,H^\top R^{-1} H, \tag{26}$$

which depends on τ only through the scalar $\gamma_\tau^2\tau/\beta_\tau^2$ and requires no Jacobian evaluation. When H and R are time-independent, $H^\top R^{-1} H$ is precomputed once and the correction costs $\mathcal{O}(N_u N_y)$ per integration step.

Remark 4.6 (Boundary behaviour). At $\tau \to 1$, $\gamma_\tau \to 0$ and $\beta_\tau \to 1$, so $G_\tau \to \boldsymbol{I}$: the interpolant and true scores coincide at the terminal time, consistent with $\bar{\boldsymbol{y}}_1 = \boldsymbol{y}$. For $\tau < 1$, the gain amplifies $\bar{S}_\tau$ along directions visible to H, with strength set by the signal-to-noise ratio $\gamma_\tau^2\tau/\beta_\tau^2$ relative to R.

5 Implementation

6 Results

We consider three test cases. Firstly, a very simple Gaussian test case where an analytical posterior is available. Secondly, a stochastic incompressible Navier Stokes test case. Lastly, a weather forecasting test case with real-world training data. Each test case serve a specific purpose which is outlined in the respective subsections.

6.1 Simple analytical test case

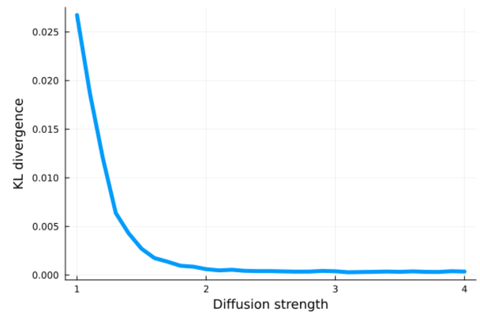
We consider the simple case:

$$\boldsymbol{u}^1 = \boldsymbol{u}^0 + \boldsymbol{\xi}, \quad \boldsymbol{\xi} \sim \mathcal{N}(0, \boldsymbol{I}), \tag{27a}$$

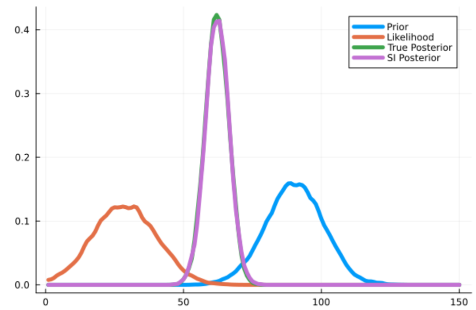
$$\boldsymbol{y}^1 = \boldsymbol{u}^1 + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(0, \boldsymbol{I}) \tag{27b}$$

Due to the simplicity, we can derive an analytical expression for the posterior. See Appendix C for more details.

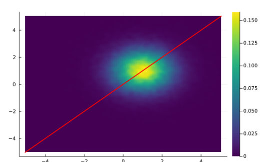
In this case, we can derive the true drift term analytically as this is a simplification of the more general case from Proposition B.9 in Chen et al. (2024). Hence, with this setup, we do not need to train any neural networks and can thereby assess the posterior sampling in isolation.



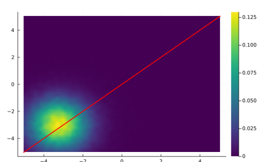
(a) KL-divergence between the SI generated posterior and the true posterior with respect to the diffusion strength.



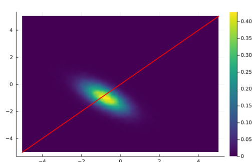
(b) Diagonal slices of the 2D distribution (red line in figure below.



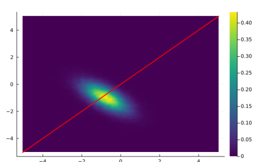
(c) Prior conditional distribution, $p(\boldsymbol{x}_1 \mid \boldsymbol{x}_0)$.



(d) Likelihood distribution centered at the observation.



(e) True analytical posterior.



(f) SI approximated posterior.

Figure 1: Caption place holder

6.2 Incompressible Navier Stokes

The second test case is an incompressible fluid flow on the torus $\mathcal{T}^2 = [0, 2\pi]^2$, governed by the 2D Navier-Stokes equations with a stochastic forcing term. The problem is formulated in terms of the vorticity field, ω:

$$\mathrm{d}\omega + \boldsymbol{v} \cdot \nabla\omega \, \mathrm{d}t = \nu \Delta \omega \, \mathrm{d}t - \alpha\omega \, \mathrm{d}t + \varepsilon \, \mathrm{d}\xi, \tag{28}$$

The velocity field $\boldsymbol{v} = \nabla^{\perp}\psi = (-\partial_y \psi, \partial_x \psi)$ is derived from the stream function $\psi(x, y)$, which satisfies $-\Delta\psi = \omega$. The stochastic term $\mathrm{d}\xi$ represents temporally white random forcing applied to selected Fourier modes.

The observation operator is linear:

$$\boldsymbol{y} = \boldsymbol{H}(\omega) + \boldsymbol{\eta}, \tag{29}$$

where $\boldsymbol{\eta} \sim \mathcal{N}(0, \sigma^2 \boldsymbol{I})$ represents observation noise with standard deviation $\sigma = 0.05$.

Table 1: Enstrophy, Energy RMSE, and KL-divergence for Navier Stokes test case.

	Vorticity RMSE		**Energy spec RMSE**		**KL-Div at points**	
	$32^2 \to 128^2$	5%	$32^2 \to 128^2$	5%	$32^2 \to 128^2$	5%
FlowDAS						
Guided FM						
Guided diffusion						
OUR METHOD						

6.3 Urban flow

Table 2: Velocity RMSE, temperature RMSE, at KL-divergence at points for the urban test case.

	Vel RMSE		**Temp RMSE**		**KL-Div at points**	
	$32^2 \to 128^2$	5%	$32^2 \to 128^2$	5%	$32^2 \to 128^2$	5%
FlowDAS						
Guided FM						
Guided diffusion						
OUR METHOD						

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A Proof of Theorem 4.1

Before we prove Theorem 4.1, we state Doob's h-transform Zhang et al.; Särkkä & Solin (2019):

Proposition A.1 (Doob's h-transform)**.** *Let $h \in C^2[\mathbb{R}^d]$ be strictly positive, and*

$$\Phi_\tau(\boldsymbol{x}) = \mathbb{E}\left[h(\boldsymbol{X}_T)|\boldsymbol{X}_\tau = \boldsymbol{x}\right], \tag{30}$$

be the solution to the Kolmogorov Backward Equation. Let $p_{\tau,\tau'}(\boldsymbol{x}, \boldsymbol{x}')$ be the Markov transition kernel of the SDE:

$$\mathrm{d}\boldsymbol{X}_\tau = \boldsymbol{b}(\boldsymbol{X}_\tau)\mathrm{d}\tau + \gamma_\tau \mathrm{d}\boldsymbol{W}_\tau, \quad \tau \in [0, T], \quad \boldsymbol{X}_0 = \boldsymbol{x}_0. \tag{31}$$

Furthermore, consider the controlled SDE:

$$\mathrm{d}\boldsymbol{X}^{\boldsymbol{u}}_\tau = \left[\boldsymbol{b}(\boldsymbol{X}^{\boldsymbol{u}}_\tau) + \gamma_\tau \boldsymbol{u}_\tau(\boldsymbol{X}^{\boldsymbol{u}}_\tau)\right]\mathrm{d}\tau + \gamma_\tau \mathrm{d}\boldsymbol{W}_\tau, \quad \tau \in [0, T], \quad \boldsymbol{X}^{\boldsymbol{u}}_0 = \boldsymbol{x}_0. \tag{32}$$

Then, if $\boldsymbol{u}_\tau(\boldsymbol{x}) = \gamma_\tau \nabla_{\boldsymbol{x}} \log \Phi_\tau(\boldsymbol{x})$, the transition kernel is:

$$p^{\boldsymbol{u}}_{\tau,\tau'}(\boldsymbol{x}, \boldsymbol{x}') = \frac{\Phi_{\tau'}(\boldsymbol{x}')}{\Phi_\tau(\boldsymbol{x})} p_{\tau,\tau'}(\boldsymbol{x}, \boldsymbol{x}'). \tag{33}$$

In particular, if $\tau = 0$, $\tau' = 1$, and $\boldsymbol{x} = \boldsymbol{x}_0$, we have:

$$p^{\boldsymbol{u}}_{0,1}(\boldsymbol{x}, \boldsymbol{x}') = \frac{h(\boldsymbol{x}')p_1(\boldsymbol{x}'|\boldsymbol{x}_0 = \boldsymbol{x})}{\Phi_0(\boldsymbol{x})}, \tag{34}$$

Where $p_1(\boldsymbol{x}'|\boldsymbol{x}_0 = \boldsymbol{x})$ is the marginal density of the process satisfying the uncontrolled SDE at time τ with initial condition $\boldsymbol{x}_0$.

A.1 Proof of Theorem 4.1

Proof. Let

$$\boldsymbol{u}(\boldsymbol{x}, \tau) = \gamma_\tau \nabla_{\boldsymbol{x}} \log \Phi^{\boldsymbol{y}}_\tau(\boldsymbol{x}, \boldsymbol{x}_0). \tag{35}$$

Then the stochastic interpolant posterior SDE is of the same form as the controlled SDE in Eq. (32).

Assuming perfect training of the drift term, we know that the marginal density, $p_1(\boldsymbol{x}'|\boldsymbol{x}_0)$, is equal to the true prior, $p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)$.

Now, define $h(\boldsymbol{x}) = p(\boldsymbol{y} \,|\, \boldsymbol{x})$. Then,

$$\Phi^{\boldsymbol{y}}_\tau(\boldsymbol{x}) = \mathbb{E}\left[p(\boldsymbol{y} \,|\, \boldsymbol{x}_1) \,|\boldsymbol{X}_\tau = \boldsymbol{x}\right], \tag{36}$$

and,

$$\Phi^{\boldsymbol{y}}_0(\boldsymbol{x}_0) = \mathbb{E}\left[p(\boldsymbol{y} \,|\, \boldsymbol{x}_1) \,|\boldsymbol{X}_0 = \boldsymbol{x}_0\right] = \int p(\boldsymbol{y} \,|\, \boldsymbol{x}_1)\, p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_1, \tag{37}$$

Then, via Doob's h-transform, we have:

$$\begin{aligned} p^{\boldsymbol{u}}_{0,1}(\boldsymbol{x}_0, \boldsymbol{x}_1) &= \frac{p(\boldsymbol{y} \,|\, \boldsymbol{x}_1)\, p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)}{\int p(\boldsymbol{y} \,|\, \boldsymbol{x}_1)\, p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_1} && \text{(38a)} \\ &= \frac{p(\boldsymbol{y} \,|\, \boldsymbol{x}_1)\, p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)}{\int p(\boldsymbol{y} \,|\, \boldsymbol{x}_1)\, p(\boldsymbol{x}_1 \,|\, \boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_1} && \text{(38b)} \\ &= p(\boldsymbol{x}_1 \,|\, \boldsymbol{y}, \boldsymbol{x}_0)\,. && \text{(38c)} \end{aligned}$$

Which was what we wanted to show.

□

B Proof of Lemma 4.2 and Lemma 4.3

Proof of Lemma 4.2. Recall the state interpolant and the observation interpolant:

$$\boldsymbol{x}_\tau = \alpha_\tau \boldsymbol{x}_0 + \beta_\tau \boldsymbol{x}_1 + \gamma_\tau \boldsymbol{W}_\tau, \tag{39}$$

$$\bar{\boldsymbol{y}}_\tau = \alpha_\tau H \boldsymbol{x}_0 + \beta_\tau \boldsymbol{y}, \tag{40}$$

where $\boldsymbol{W}_\tau = \sqrt{\tau}\,\boldsymbol{z}$ with $\boldsymbol{z} \sim \mathcal{N}(0, \boldsymbol{I})$, and $\boldsymbol{y} = H\boldsymbol{x}_1 + \boldsymbol{\eta}$ with $\boldsymbol{\eta} \sim \mathcal{N}(0, R)$ independent of $\boldsymbol{z}$. Substituting $\boldsymbol{y} = H\boldsymbol{x}_1 + \boldsymbol{\eta}$ into (40):

$$\bar{\boldsymbol{y}}_\tau = H(\alpha_\tau \boldsymbol{x}_0 + \beta_\tau \boldsymbol{x}_1) + \beta_\tau \boldsymbol{\eta}. \tag{41}$$

From (39), $\alpha_\tau \boldsymbol{x}_0 + \beta_\tau \boldsymbol{x}_1 = \boldsymbol{x}_\tau - \gamma_\tau \boldsymbol{W}_\tau$, giving

$$\bar{\boldsymbol{y}}_\tau = H\boldsymbol{x}_\tau + \underbrace{\beta_\tau \boldsymbol{\eta} - \gamma_\tau H \boldsymbol{W}_\tau}_{=\bar{\boldsymbol{\eta}}_\tau}. \tag{42}$$

Since $\boldsymbol{\eta}$ is independent of both $\boldsymbol{x}_1$ and $\boldsymbol{z}$, we have $\boldsymbol{\eta} \perp (\boldsymbol{W}_\tau, \boldsymbol{x}_\tau) \mid \boldsymbol{x}_0$. Therefore $\boldsymbol{\eta} \perp \boldsymbol{x}_\tau \mid \boldsymbol{x}_0$, and

$$\mathbb{E}[\boldsymbol{\eta} \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0] = 0, \qquad \mathrm{Cov}(\boldsymbol{\eta} \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0) = R. \tag{43}$$

Using (42) and the independence of $\boldsymbol{\eta}$:

$$\begin{aligned}\mathbb{E}[\bar{\boldsymbol{y}}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0] &= H\boldsymbol{x}_\tau + \beta_\tau \mathbb{E}[\boldsymbol{\eta} \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0] - \gamma_\tau H\, \mathbb{E}[\boldsymbol{W}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0] \\ &= H\boldsymbol{x}_\tau - \gamma_\tau H\, \mathbb{E}[\boldsymbol{W}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0].\end{aligned} \tag{44}$$

The bias term is $-\gamma_\tau H\, \mathbb{E}[\boldsymbol{W}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0]$, which is nonzero whenever $\gamma_\tau > 0$ and $\mathbb{E}[\boldsymbol{W}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0] \neq 0$.

Since $\boldsymbol{\eta} \perp (\boldsymbol{W}_\tau, \boldsymbol{x}_\tau) \mid \boldsymbol{x}_0$, the cross-terms between $\boldsymbol{\eta}$ and $\boldsymbol{W}_\tau$ vanish upon conditioning on $(\boldsymbol{x}_\tau, \boldsymbol{x}_0)$:

$$\begin{aligned}\mathrm{Cov}(\bar{\boldsymbol{\eta}}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0) &= \beta_\tau^2\, \mathrm{Cov}(\boldsymbol{\eta} \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0) + \gamma_\tau^2\, H\, \mathrm{Cov}(\boldsymbol{W}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0)\, H^\top \\ &= \beta_\tau^2\, R + \gamma_\tau^2\, H\, \mathrm{Cov}(\boldsymbol{W}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0)\, H^\top.\end{aligned} \tag{45}$$

□

Proof of Lemma 4.3. We write $\boldsymbol{W}_\tau = \sqrt{\tau}\,\boldsymbol{z}$ and work with the standard normal $\boldsymbol{z}$. All expectations below are conditioned on $(\boldsymbol{X}_\tau = \boldsymbol{x}, \boldsymbol{X}_0 = \boldsymbol{x}_0)$.

Conditional mean. Via Stein's formula Albergo et al. (2023); Chen et al. (2024), the conditional expectation of $\boldsymbol{z}$ relates to the score:

$$\mathbb{E}[\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{x}_0] = -\gamma_\tau \sqrt{\tau}\, \nabla_{\boldsymbol{x}} \log p_\tau(\boldsymbol{x} \,|\, \boldsymbol{x}_0)\,. \tag{46}$$

Therefore,

$$\mathbb{E}[\boldsymbol{W}_\tau \mid \boldsymbol{x}, \boldsymbol{x}_0] = \sqrt{\tau}\, \mathbb{E}[\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{x}_0] = -\gamma_\tau \tau\, \nabla_{\boldsymbol{x}} \log p_\tau(\boldsymbol{x} \,|\, \boldsymbol{x}_0)\,. \tag{47}$$

Using the relation from Chen et al. (2024),

$$\nabla_{\boldsymbol{x}} \log p_\tau(\boldsymbol{x} \,|\, \boldsymbol{x}_0) = A_\tau \left[\beta_\tau \boldsymbol{b}_\theta(\boldsymbol{x}, \boldsymbol{x}_0, \tau) - c_\tau(\boldsymbol{x}, \boldsymbol{x}_0)\right], \tag{48}$$

and substituting into (47), we obtain:

$$\mathbb{E}[\boldsymbol{W}_\tau \mid \boldsymbol{x}, \boldsymbol{x}_0] = -\gamma_\tau \tau A_\tau (\beta_\tau \boldsymbol{b}_\theta(\boldsymbol{x}, \boldsymbol{x}_0, \tau) - c_\tau(\boldsymbol{x}, \boldsymbol{x}_0)). \tag{49}$$

Conditional covariance. The interpolant can be written as $\boldsymbol{x}_\tau = \boldsymbol{m}_\tau + \sigma_\tau \boldsymbol{z}$, where $\boldsymbol{m}_\tau = \alpha_\tau \boldsymbol{x}_0 + \beta_\tau \boldsymbol{x}_1$ and $\sigma_\tau = \gamma_\tau \sqrt{\tau}$. For fixed $\boldsymbol{x}_0$, the marginal density of $\boldsymbol{x}_\tau$ is a Gaussian mixture (over $\boldsymbol{x}_1$) with per-component variance $\sigma_\tau^2 \boldsymbol{I}$. The second-order Tweedie identity gives the conditional covariance of $\boldsymbol{m}_\tau$:

$$\mathrm{Cov}\left(\boldsymbol{m}_\tau \mid \boldsymbol{x}, \boldsymbol{x}_0\right) = \sigma_\tau^2 \boldsymbol{I} + \sigma_\tau^4 \nabla_{\boldsymbol{x}}^2 \log p_\tau(\boldsymbol{x} \,|\, \boldsymbol{x}_0)\,. \tag{50}$$

Since $\boldsymbol{z} = (\boldsymbol{x}_\tau - \boldsymbol{m}_\tau)/\sigma_\tau$, we have:

$$\mathrm{Cov}\left(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{x}_0\right) = \frac{1}{\sigma_\tau^2} \mathrm{Cov}\left(\boldsymbol{m}_\tau \mid \boldsymbol{x}, \boldsymbol{x}_0\right) \tag{51}$$

$$= \boldsymbol{I} + \sigma_\tau^2 \nabla_{\boldsymbol{x}}^2 \log p_\tau(\boldsymbol{x} \,|\, \boldsymbol{x}_0)\,. \tag{52}$$

Then, since $\boldsymbol{W}_\tau = \sqrt{\tau}\, \boldsymbol{z}$:

$$\mathrm{Cov}\left(\boldsymbol{W}_\tau \mid \boldsymbol{x}, \boldsymbol{x}_0\right) = \tau \left(\boldsymbol{I} + \gamma_\tau^2 \tau \nabla_{\boldsymbol{x}}^2 \log p_\tau(\boldsymbol{x} \,|\, \boldsymbol{x}_0)\right). \tag{53}$$

Taking the Jacobian of the score with respect to $\boldsymbol{x}$:

$$\nabla_{\boldsymbol{x}}^2 \log p_\tau(\boldsymbol{x} \,|\, \boldsymbol{x}_0) = A_\tau \left[\beta_\tau J_{\boldsymbol{b}_\theta}(\boldsymbol{x}, \boldsymbol{x}_0, \tau) - \dot{\beta}_\tau \boldsymbol{I}\right], \tag{54}$$

where we used $\nabla_{\boldsymbol{x}} c_\tau(\boldsymbol{x}, \boldsymbol{x}_0) = \dot{\beta}_\tau \boldsymbol{I}$. Substituting into (53) yields:

$$\mathrm{Cov}\left(\boldsymbol{W}_\tau \mid \boldsymbol{x}, \boldsymbol{x}_0\right) = \tau \boldsymbol{I} + \gamma_\tau^2 \tau^2 A_\tau \left[\beta_\tau J_{\boldsymbol{b}_\theta}(\boldsymbol{x}, \boldsymbol{x}_0, \tau) - \dot{\beta}_\tau \boldsymbol{I}\right]. \tag{55}$$

□

C Simple test case

The analytical drift term for the SI SDE that transforms $\boldsymbol{x}_0$ samples to $\boldsymbol{x}_1$ samples is:

$$\boldsymbol{b}(\boldsymbol{x}, \boldsymbol{x}_0, \tau) = \dot{\alpha}_\tau \boldsymbol{x}_0 + \dot{\beta}_\tau \bar{\boldsymbol{\mu}}(\boldsymbol{x}_0) + (\beta_\tau \dot{\beta}_\tau \boldsymbol{I} + \tau \gamma_\tau \dot{\gamma}_\tau \boldsymbol{I}) \bar{\boldsymbol{\Sigma}}^{-1} (\boldsymbol{x} - \bar{\boldsymbol{\mu}}(\boldsymbol{x}_0)), \tag{56}$$

where

$$\bar{\boldsymbol{\mu}}(\boldsymbol{x}_0) = \alpha_\tau \boldsymbol{x}_0 + \beta_\tau \boldsymbol{x}_0, \quad \bar{\boldsymbol{\Sigma}} = \beta_\tau^2 \boldsymbol{I} + \tau \gamma_\tau^2 \boldsymbol{I}. \tag{57}$$

The true posterior is:

$$p\big(\boldsymbol{u}^1 \,\big|\, \boldsymbol{y}^1\big) = \mathcal{N}\left(\boldsymbol{u}^0 + K(\boldsymbol{y}^1 - \boldsymbol{u}^0), \boldsymbol{I} - K\right), \tag{58}$$

where

$$K = H^\top (H H^\top + \boldsymbol{I})^{-1} = 0.5 \boldsymbol{I}. \tag{59}$$

D Proofs for the multiplicative correction

We first establish a tractable form of the true-likelihood mean, and then a residual identity relating the two Gaussian likelihoods, before proving Theorem 4.4.

Lemma D.1 (Tractable true-likelihood mean)**.** *Let the state interpolant be $\boldsymbol{x}_\tau = \alpha_\tau \boldsymbol{x}_0 + \beta_\tau \boldsymbol{x}_1 + \gamma_\tau \boldsymbol{W}_\tau$ with $\beta_\tau > 0$, and let $\boldsymbol{H}$ be linear with matrix H. Then*

$$\mathbb{E}[\boldsymbol{x}_1 \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0] = \frac{1}{\beta_\tau}\left(\boldsymbol{x}_\tau - \alpha_\tau \boldsymbol{x}_0 - \gamma_\tau \mu_\tau^{\boldsymbol{W}}(\boldsymbol{x}_\tau, \boldsymbol{x}_0)\right), \tag{60}$$

and consequently

$$\mu_\tau(\boldsymbol{x}_\tau, \boldsymbol{x}_0) = \frac{H}{\beta_\tau}\left(\boldsymbol{x}_\tau - \alpha_\tau \boldsymbol{x}_0 - \gamma_\tau \mu_\tau^{\boldsymbol{W}}(\boldsymbol{x}_\tau, \boldsymbol{x}_0)\right), \tag{61}$$

which depends on $\boldsymbol{b}_\theta$ only through the conditional Wiener mean $\mu_\tau^{\boldsymbol{W}}$ of Lemma 4.3, not through its Jacobian.

Proof. Solving the state interpolant for $\boldsymbol{x}_1$ and taking conditional expectation with respect to $(\boldsymbol{x}_\tau, \boldsymbol{x}_0)$ gives (60) upon identifying $\mathbb{E}[\boldsymbol{W}_\tau \mid \boldsymbol{x}_\tau, \boldsymbol{x}_0] = \mu_\tau^{\boldsymbol{W}}$. Applying H yields (61). □

Lemma D.2 (Residual identity)**.** *Under the assumptions of Lemma 4.2 and with μ_τ defined by* (61),

$$\bar{\boldsymbol{y}}_\tau - \bar{\mu}_\tau = \beta_\tau(\boldsymbol{y} - \mu_\tau), \tag{62}$$

and the gradients of the two means satisfy

$$\nabla_{\boldsymbol{x}_\tau}\bar{\mu}_\tau = \beta_\tau \nabla_{\boldsymbol{x}_\tau}\mu_\tau, \qquad \nabla_{\boldsymbol{x}_\tau}\mu_\tau = \frac{H}{\beta_\tau \tau}\,\Sigma_\tau^{\boldsymbol{W}}(\boldsymbol{x}_\tau, \boldsymbol{x}_0). \tag{63}$$

Proof. From Lemma 4.2, $\bar{\mu}_\tau = H\boldsymbol{x}_\tau - \gamma_\tau H\mu_\tau^{\boldsymbol{W}}$, and from (13), $\bar{\boldsymbol{y}}_\tau = \alpha_\tau H\boldsymbol{x}_0 + \beta_\tau \boldsymbol{y}$. Subtracting and using (61) yields (62). Differentiating (62) in $\boldsymbol{x}_\tau$ gives the first equality in (63). For the second, differentiating (61),

$$\nabla_{\boldsymbol{x}_\tau}\mu_\tau = \frac{H}{\beta_\tau}\big(\boldsymbol{I} - \gamma_\tau \nabla_{\boldsymbol{x}_\tau}\mu_\tau^{\boldsymbol{W}}\big)\,, \tag{64}$$

and from Lemma 4.3 together with $\nabla_{\boldsymbol{x}} c_\tau = \dot{\beta}_\tau \boldsymbol{I}$,

$$\nabla_{\boldsymbol{x}_\tau}\mu_\tau^{\boldsymbol{W}} = -\gamma_\tau \tau A_\tau\big(\beta_\tau J_{\boldsymbol{b}_\theta} - \dot{\beta}_\tau \boldsymbol{I}\big)\,, \tag{65}$$

so that $\Sigma_\tau^{\boldsymbol{W}} = \tau\big(\boldsymbol{I} - \gamma_\tau \nabla_{\boldsymbol{x}_\tau}\mu_\tau^{\boldsymbol{W}}\big)$ by Eq. (21). Substituting into (64) completes the proof. □

Proof of Theorem 4.4. Under (22) and using Lemma D.2,

$$\bar{S}_\tau = (\nabla_{\boldsymbol{x}_\tau}\bar{\mu}_\tau)^\top \bar{\Sigma}_\tau^{-1}(\bar{\boldsymbol{y}}_\tau - \bar{\mu}_\tau) = \frac{\beta_\tau}{\tau}\,\Sigma_\tau^{\boldsymbol{W}}\,H^\top \bar{\Sigma}_\tau^{-1}\,(\boldsymbol{y} - \mu_\tau), \tag{66}$$

and under (23),

$$S_\tau = (\nabla_{\boldsymbol{x}_\tau}\mu_\tau)^\top R^{-1}\,(\boldsymbol{y} - \mu_\tau) = \frac{1}{\beta_\tau \tau}\,\Sigma_\tau^{\boldsymbol{W}}\,H^\top R^{-1}\,(\boldsymbol{y} - \mu_\tau). \tag{67}$$

By Lemma 4.2, $\bar{\Sigma}_\tau = \beta_\tau^2 R + \gamma_\tau^2 H \Sigma_\tau^{\boldsymbol{W}} H^\top$, and the Woodbury identity gives

$$\bar{\Sigma}_\tau^{-1} = \tfrac{1}{\beta_\tau^2}\,R^{-1} - \tfrac{1}{\beta_\tau^4}\,R^{-1}H\,T_\tau\,H^\top R^{-1}, \qquad T_\tau := \big(\gamma_\tau^{-2}(\Sigma_\tau^{\boldsymbol{W}})^{-1} + \beta_\tau^{-2}\,H^\top R^{-1} H\big)^{-1}. \tag{68}$$

Left-multiplying by $H^\top$ and using $T_\tau^{-1} - \beta_\tau^{-2}\,H^\top R^{-1} H = \gamma_\tau^{-2}(\Sigma_\tau^{\boldsymbol{W}})^{-1}$,

$$H^\top \bar{\Sigma}_\tau^{-1} = \tfrac{1}{\beta_\tau^2 \gamma_\tau^2}\,(\Sigma_\tau^{\boldsymbol{W}})^{-1}\,T_\tau\,H^\top R^{-1}. \tag{69}$$

Substituting (69) into (66),

$$\bar{S}_\tau = \tfrac{1}{\beta_\tau \tau \gamma_\tau^2}\,T_\tau\,H^\top R^{-1}\,(\boldsymbol{y} - \mu_\tau). \tag{70}$$

Comparing (67) and (70), the multiplicative factor relating them is

$$G_\tau = \gamma_\tau^2\,\Sigma_\tau^{\boldsymbol{W}}\,T_\tau^{-1} = \gamma_\tau^2\,\Sigma_\tau^{\boldsymbol{W}}\big(\gamma_\tau^{-2}(\Sigma_\tau^{\boldsymbol{W}})^{-1} + \beta_\tau^{-2}\,H^\top R^{-1} H\big) = \boldsymbol{I} + \tfrac{\gamma_\tau^2}{\beta_\tau^2}\,\Sigma_\tau^{\boldsymbol{W}}\,H^\top R^{-1} H, \tag{71}$$

and indeed $G_\tau \bar{S}_\tau = S_\tau$. $\operatorname{rank}(G_\tau - \boldsymbol{I}) \le \operatorname{rank}(H) \le N_y$, and $(G_\tau - \boldsymbol{I})v = 0$ for $v \in \ker H$. □